

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2019- 2020 (Odd Semester)

Innovative Practices Description

UNIT-I (ELECTRO-PHYSIOLOGY AND BIO-POTENTIAL RECORDING)

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8073 Medical Electronics

Name of the Faculty member: Mr.A.Rameshbabu

Name of the Topic: Bio- potential

Name of the Innovative Practice: Animation Video

Date & Duration: 03.07.2019 & 10 minutes

ICT Tool Used: Desktop, projector

Description:

Animation is a brilliant and innovative new way to encourage students to communicate stories, ideas and concepts in a creative and original way. It can be particularly useful as a tool to encourage the creativity of students. Animated videos are a one stop solution that can grab the attention of the students. With its entertaining visuals, it can clearly comprehend the concept. Since it is an analogy to cartoons, students will get hooked to it. By using animation videos, students learn analytical thinking as to how to present a scenario.

Goals (Learning Outcomes):

- The students will be able understand about the chemical reactions of sodium and potassium ions.
- The students will be able to explain the application of Bio - potentials.

Use of appropriate methods:

Justification for choosing the topic using animation video activity:

Bio potential is an electric signal created in our body. The sodium and potassium ion movement creates electric signals. Animated video helps to show how the ions movement in our body creates bio electric signals.

Effective presentation

Details of Implementation:

The suitable animated video for Bio-potential was displayed to the students and explained about how the ions movement in our body creates bio electric signals.

Innovative Teaching Method Execution	
Bio- potential - Animated video	
	

Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 85% of students answered correctly.

Reflective Critique:

- **Benefits:**

The students are more interested to watch animated video for Bio - potential. It helps the students to know how the ions movement in our body creates bio electric signals. It helps to differentiate the different types of ions movement in our body.

- **Challenges:**

It takes some time to choose relevant videos for the Bio - potential topic.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	1	-	-	-	-	-	1	1	1	-	-

CO1: The students will be able to explain about the physiological parameters and recording methods.

Reference:

1. Khandpur, R.S., “Handbook of Biomedical Instrumentation”, TATA Mc Graw-Hill, New Delhi, 2003.
2. <https://www.youtube.com/watch?v=yYFF4pvliBk>
3. [https:// www.khanacademy.org](https://www.khanacademy.org)

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UNIT – II (BIO-CHEMICAL AND NON-ELECTRICAL PARAMETER MEASUREMENT)

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8073 Medical Electronics

Name of the Faculty member: Mr.A.Rameshbabu

Name of the Topic: Blood Cell Counters

Date & Duration: 05.08.2019 & 1 minute

ICT Tool Used:-

Description:

One minute paper:

The Minute Paper is a very commonly used classroom assessment technique. It really does take about a minute and, while usually used at the end of class, it can be used at the end of any topic discussion. Its major advantage is that it provides rapid feedback on whether the professor's main idea and what the students perceived as the main idea are the same. Additionally, by asking students to add a question at the end, this assessment becomes an integrative task. Students must first organize their thinking to rank the major points and then decide upon a significant question. Sometimes, instead of asking for the main point, a professor may wish to probe for the most disturbing or most surprising item. It is thus a very adaptable tool.

Goals (Learning Outcomes):

- Students will able to Understand about Blood Cells and their functions.
- Students will able to explain about different types of Blood Cell Counting methods.

Use of appropriate methods:

Justification for choosing the One minute paper Activity:

This Activity will take only one minute time duration. It is easy to recall all point in that particular topic. Blood Cell Counter topic is an important topic in unit 2 that was the reason behind to choose this topic for One Minute Paper activity.

Time duration & Challenges:

Initially, I have planned this activity for only one minute but students were taken one minute to become ready for this activity. So, that I have extent the activity for Two minutes. Next time I want to give prior information about this activity to my students to reduce the time duration.

Effective presentation

Details of the Implementation:

The students confused about the different types of Blood Cells & Counting methods which was explained in the class room. After that, I have chosen One Minute Paper as an Innovative practice method for better understanding about Blood Cell Counter. The Students eagerly participated in that activity and then they acquired the concept of Blood cells and their functions.

Innovative Teaching Method Execution

Blood Cell Counters :- One minute paper



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- After conducting this activity I found that many students understand this topic.
- The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correctly.

Reflective Critique:

- **Benefits:**

Minute papers are an effective way of involving all students in class simultaneously. It ensures equal participation of each and every class member, including anyone who may be too shy or fearful to participate orally. It can promote class attendance and attentiveness. It can improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

- **Challenges:**

After teaching this particular topic students asked some time to prepare. So, I gave few minutes to prepare which takes more time to implement this activity than planned.

Changes required for future (if required): Nil

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO2	1	1	-	-	-	-	-	1	1	1	-	-

CO2: The students will be able to analyze the bio-chemicals and various physiological information.

References:

- 1) Khandpur, R.S., “Handbook of Biomedical Instrumentation”, TATA Mc Graw-Hill, New Delhi, 2003.
- 2) <https://oncourseworkshop.com/self-awareness/one-minute-paper/>
- 3) https://www.youtube.com/watch?v=R_nf0J3i1w

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Innovative Practices Description

UNIT – III (ASSIST DEVICES)

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8073 Medical Electronics

Name of the Faculty member: Mr.A.Rameshbabu

Name of the Topic: Magnetic Resonance Imaging Systems

Name of the Innovative Practice: Mind maps

Date & Time: 19.08.2019 & 15 Minutes

ICT Tool Used: Black Board

Description:

Mind Map Technique:

A mind map is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

A mind map is a diagram for representing tasks, words, concepts, or items linked to and arranged around a subject using a non-linear graphical layout that allows the student to build an intuitive framework around a central concept.

Goals (Learning Outcomes):

1. Students will be able to understand about Magnetic Resonance Imaging Systems and their working functions.
2. Students will be able to explain about different types of Real time applications in MRI Systems.

Use of appropriate methods:

Justification for choosing the topic using mind map activity:

Mind map activity gives the graphical representation of MRI systems. It differentiates the MRI systems and Ultrasound Imaging systems clearly which will be helpful for understanding the concepts of imaging systems to the students.

Effective presentation

Details of the Implementation:

The students confused about the different types of Imaging methods which was explained in the class room. After that, I have chosen Mind map as an Innovative practice method for better understanding about Magnetic Resonance Imaging Systems. The Students eagerly participated in that activity and then they acquired the concept of MRI imaging system functions.

Innovative Teaching Method Execution

Magnetic Resonance Imaging Systems:- Mind maps



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- Every students got equal opportunity to come forward to take part in this activity
- The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 80% of students answered correctly.

Reflective Critique:

• **Benefits:**

Making a mind map is an excellent way for students to be able to sort through your thoughts and ideas. This activity allows the students to quickly generate creative and even unique ideas in less time. with the help of this activity the students can able to easily differentiate between different building blocks of MRI.

• **Challenges:**

Initially, I have planned the activity for 10 minutes but some students not aware of this activity. So, that I have explained the activity and then some of the students hesitated for doing that. I encouraged the students which it takes another 5 minutes for completing the activity.

Changes required for future (if required): Planned to increase the Time Duration

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	1	1	-	-	-	-	-	1	-	1	-	-

CO3: The students will be able to describe various assist devices used in hospitals.

References:

- 1) Khandpur, R.S., “Handbook of Biomedical Instrumentation”, TATA Mc Graw-Hill, New Delhi, 2003
- 2) <https://www.qualitymatters.org/qa-resources/resource-center/articles-resources/mind-maps>

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Innovative Practices Description

UNIT-IV(PHYSICAL MEDICINE AND BIOTELEMETRY)

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8073 Medical Electronics

Name of the Faculty member: Mr.A.Rameshbabu

Name of the Topic: Diathermy

Name of the Innovative Practice: Animation Video

Date & Duration: 30.08.2019 & 10 minutes

ICT Tool Used: Desktop, projector

Description:

Animation is a brilliant and innovative new way to encourage students to communicate stories, ideas and concepts in a creative and original way. It can be particularly useful as a tool to encourage the creativity of students. Animated videos are a one stop solution that can grab the attention of the students. With its entertaining visuals, it can clearly comprehend the concept. Since it is an analogy to cartoons, students will get hooked to it. By using animation videos, students learn analytical thinking as to how to present a scenario.

Goals (Learning Outcomes):

- The students will be able to easily identify the different types of diathermy methods used for medical treatment.
- The students will be able to explain the application of diathermy process.

Use of appropriate methods:

Justification for choosing the topic using animation video activity:

Diathermy means “through heating” or producing deep heating directly in the tissues of the body. The heat is applied with the help of electrodes. Animated video helps to show how the heat is developed in the electrode and the same is applied to the body to heal the wounds.

Effective presentation

Details of Implementation:

The suitable animated video for diathermy was displayed to the students and explained about the different types of diathermy and their medical applications.

Innovative Teaching Method Execution

Diathermy - Animated video



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 85% of students answered correctly.

Reflective Critique:

- **Benefits:**

The students are more interested to watch animated video, apart from theory concept. Animation videos for diathermy help the students to know how the diathermy helps to heal the tissues and wounds. It helps to differentiate the different types of diathermy and their related procedure on applying to patients.

- **Challenges:**

It takes some time to choose relevant videos for the diathermy topic.

Changes required for future (if required):

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO4	1	1	-	-	-	-	-	1	1	1	-	-

CO4: The students will be able to explain the equipment used for physical medicine and the various recently developed diagnostic and therapeutic techniques

Reference:

1. Khandpur, R.S., “Handbook of Biomedical Instrumentation”, TATA Mc Graw-Hill, New Delhi, 2003.
2. <https://www.youtube.com/watch?v=iqDt2NvQP1Y>

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UNIT V (RECENT TRENDS IN MEDICAL INSTRUMENTATION)

Degree, Semester& Branch: V Semester B.E. ECE A

Course Code & Title: EC8073 & Medical Electronics

Name of the Faculty member: Mr.A.Rameshbabu

Name of the Topic: Lab on a chip

Name of the Innovative Practice: Brainstorming

Date & Time: 03.10.19 & 30 Minutes

ICT Tool Used: - Desktop computer, projector

Description:

Brainstorming is a group creativity technique by which efforts are made to find a conclusion for a specific problem by gathering a list of ideas spontaneously contributed by its members. All the ideas are noted down without criticism and after the brainstorming session the ideas are evaluated.

Goals (Learning Outcomes):

- The students will be able to develop their creative thinking and problem solving skills.
- The students will be able to know about modern technologies used in medical laboratories.

Use of appropriate methods:

Justification for choosing the topic using Brainstorming activity:

Lab on a chip is miniaturized integrated circuit which consists of sensors, transducers, amplifiers which can measure or test pH level, blood pressure, blood flow rate simultaneously and give the output etc., the students already learned this concept in previous chapter. So I asked them to prepare and share their ideas.

Effective presentation

Details of Implementation:

The process begins with a well-defined topic. Students are divided into groups. Each participant in a group creates an individual brainstorm around the topic. Each group members need to present their views on stage. All the ideas are then merged into one large idea map. During this consolidation phase the participants may discover a common understanding of the issues as they share the meanings behind their ideas. As the sharing takes place, new ideas may arise by the association. Those ideas are added to the map as well. Then ideas are generated on both individual and group levels.

Innovative Teaching Method Execution

Lab on a chip - Brainstorming



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- The success of the activity was evaluated by asking the same question in Internal Assessment test III – Around 80% of students answered correctly.
- Slow learners came forward to explain their ideas.

Reflective Critique:

- **Benefits:**

Brainstorming builds involvement, commitment, loyalty, and enthusiasm. Participating in the sessions stimulates and unlocks people's creative talents. Brainstorming also builds self-esteem because people are being asked for their participation and their ideas.

- **Challenges:**

For each student i allotted for 5 minutes to present their ideas on the stage. But few students take more than the time allotted.

Changes required for future (if required): Plan to deliver detailed explanation about this activity

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO5	1	1	-	-	-	-	-	1	-	1	-	-

CO5: The students will be able to apply the concepts of medical Instrumentation in recent technology (Radio pill, Telemedicine, Endomicroscopy unit)

References:

1. Khandpur, R.S., "Handbook of Biomedical Instrumentation", TATA Mc Graw-Hill, New Delhi, 2003.
2. <https://www.youtube.com/watch?v=YXZamW4-Ysk>
3. <https://technologyadvice.com/blog/marketing/brainstorming-activities-inspire-content-marketing-team/>